

Trigonometric Identities — Problem Set

Warm-up

1. Use the compound angle formula for $\tan(A - B)$ to simplify $\frac{\tan 5\theta - \tan 2\theta}{1 + \tan 5\theta \tan 2\theta}$.
2. Expand $\sin(2x + x)$ using the sine sum formula.
3. If $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$, find the exact value of $\sin 2\theta$.
4. State the three different forms of the double angle formula for $\cos 2A$.
5. Simplify the expression $2 \sin 15^\circ \cos 15^\circ$ using a double angle identity.
6. Given $\tan A = 2$, find the value of $\tan 2A$.
7. Rewrite $1 - 2 \sin^2(3x)$ as a single trigonometric function.
8. Simplify $\frac{\sin 2\theta}{2 \sin \theta}$.
9. Express $\cos^2 \alpha - \sin^2 \alpha$ in terms of 2α .
10. Use the identity for $\sin(A - B)$ to find the exact value of $\sin 15^\circ$ (take $A = 45^\circ, B = 30^\circ$).

Skill-building

1. Show that $\frac{\sin 2x}{1 + \cos 2x} = \tan x$.
2. Prove the identity $\frac{\cos A - \cos 3A}{\sin 3A - \sin A} = \tan 2A$.
3. Solve $2 \sin \theta \cos \theta = \frac{1}{2}$ for $0 \leq \theta \leq \pi$.
4. Given $\cos 2\theta = \frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$, find the exact value of $\cos \theta$.
5. Prove that $\cos^4 \theta - \sin^4 \theta = \cos 2\theta$.
6. Simplify the expression $\frac{\sin 3\alpha}{\sin \alpha} - \frac{\cos 3\alpha}{\cos \alpha}$ to a constant integer.
7. Find all solutions to $\cos 2x + 3 \cos x + 2 = 0$ in the domain $0 \leq x \leq 2\pi$.
8. Prove that $\csc 2\theta + \cot 2\theta = \cot \theta$.
9. If $\tan x = t$, show that $\sin 2x = \frac{2t}{1+t^2}$.
10. Use the product-to-sum identities to express $\sin 5x \cos 3x$ as a sum of two sine functions.

Easier Exam Questions

1. (Hornsby Girls 2021 Q5) Which expression is equivalent to $\frac{\tan 3x - \tan x}{1 + \tan 3x \tan x}$?

- (A) $\tan 2x$
- (B) $\tan 4x$
- (C) $\tan 6x$
- (D) $\frac{\tan 3x - 1}{1 + \tan 3x}$

2. (Girraween 2024 Q4)

Which expression is equivalent to $\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x}$

- A. $\cot 2x$
- B. $\tan 2x$
- C. $\frac{\tan 2x}{1 + \tan 5x \tan 3x}$
- D. $\cot x$

3. (Baulkham Hills 2024 Q5)

Which of the following is equivalent to $\frac{1 - \cos 2x}{\sin 2x}$?

- A. $1 - \cot 2x$
- B. 1
- C. $\cot x$
- D. $\tan x$

4. (Hornsby Girls 2021 Q4) The angle θ satisfies $\sin \theta = \frac{12}{13}$ for $\frac{\pi}{2} < \theta < \pi$.

The value of $\cos 2\theta$ is...

- (A) $\frac{10}{13}$
- (B) $-\frac{10}{13}$
- (C) $\frac{119}{169}$
- (D) $-\frac{119}{169}$

5. (Presbyterian 2023 Q12b)

Simplify $\frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x}$

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6. (Gosford 2020 Q9c)

Show that $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x} = \tan x$.

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Harder Exam Questions

1. (Blacktown Boys 2022 Q22b)

$$2 \cos^2 2x - \cos 2x - 1 = 0, 0 \leq x \leq \pi$$

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2. (Blacktown Boys 2022 Q24)

Prove that $\tan \theta (1 + \tan \theta) + (1 + \cot \theta) = \frac{\tan \theta + 1}{\sin \theta \cos \theta}$

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3. (Girraween 2024 Q4 a, b, d & e)
- (a) Using the expansion of $\cos(A + B)$, find the exact value of $\sec 105^\circ$ **3**
- (b)
- i. Prove that $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$ **2**
- ii. Hence show the exact value of $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$ **2**
- (d) Show that $\sin(A - 2B) \cos(A + 2B) = \frac{1}{2}(\sin 2A - \sin 4B)$ **2** Note: Sine is an odd function and Cosine is an even function.
- (e) Prove the identity $\frac{2 \sin^3 A + 2 \cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$ **2**
4. (Baulkham Hills 2024 Q11c)
- (i) Show that $\cot \theta - \cot 2\theta = \operatorname{cosec} 2\theta$. **2**
- (ii) Hence find the exact value of $\cot 15^\circ$. **1**
5. (Sydney Girls 2021 Q4d)
- Prove that: $\tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = 2 \sec 2\theta$ **3**
6. (Sydney Girls 2020 Q4c)
- Suppose that x is a real number and that m and n are positive integers.
- (i) Express $\sin mx \sin \frac{x}{2}$ as a difference of cosine functions. **1**
- (ii) Hence, show that **2**
- $$\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\cos \frac{x}{2} - \cos\left(n + \frac{1}{2}\right)x}{2 \sin \frac{x}{2}}.$$
7. (Sydney Girls 2022 Q2b)
- (i) Prove that $(\sin \alpha - \cos \alpha)^2 = 1 - \sin 2\alpha$. **2**
- (ii) Hence, or otherwise, find the exact value of $\sin 75^\circ - \cos 75^\circ$. Leave your answer with a rational denominator. **2**
8. (Normanhurst Boys 2019 Q15 a,b& c)
- (a) Find the exact value of $(\sin 22.5^\circ + \cos 22.5^\circ)^2$ **2**
- (b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$. **4**
- Find the exact value of $\tan(x + y)$.
- (c) Prove that $\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} = 4 \cos 2\theta$ **3**
9. (Caringbah 2025 Q8b)
- (a) Find the exact value of $(\sin 22.5^\circ + \cos 22.5^\circ)^2$ **2**
- (b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$. **4**
- Find the exact value of $\tan(x + y)$.
- (c) Prove that $\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} = 4 \cos 2\theta$ **3**

10. (Caringbah 2025 Q10c)

Solve

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$$\cos^2 x + \cos^2(2x) + \cos^2(3x) = 1, 0 \leq x \leq 2\pi$$